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Research Statement

LLM Fine-Tuning for Mechanics

My first research project explored language-model fine-tuning for question answering in mechanics of materials. I examined the organization of existing instruction datasets and prepared question–answer pairs and concept explanations from textbooks and technical literature. Using these data, I fine-tuned Qwen2.5-7B with LoRA and released two instruction datasets and two model checkpoints on Hugging Face. Preparing the training examples taught me to consider how the structure and content of a dataset should reflect the task a model is expected to perform. This experience gave me an initial understanding of the relationship between data preparation and model training, which I continued to explore in subsequent projects on predicting physical fields.

ConvLSTM Prediction of Battery Material Fields

In a subsequent project on battery materials, I automated MATLAB–COMSOL simulations to generate time-aligned image sequences of lithium concentration and von Mises stress. I trained separate conditional ConvLSTM models in PyTorch to predict the evolution of these fields. During autoregressive prediction, the models used their own outputs as inputs for subsequent steps, allowing errors to accumulate over time. To reduce the mismatch between training and deployment, I used scheduled sampling to gradually introduce model-generated inputs during training. I evaluated multistep predictions against COMSOL reference sequences to examine how errors developed over the prediction horizon. This project led me to consider how both training strategies and evaluation procedures should reflect the conditions under which a model will actually be used.

PINNs for Shock Capturing

I next investigated physics-informed neural networks (PINNs) for a one-dimensional inviscid Burgers problem involving shock formation. Using PyTorch and automatic differentiation, I constructed a training loss incorporating the PDE residual, initial conditions, and boundary conditions. I then introduced an artificial-viscosity term with a fixed coefficient into the PDE residual and compared the resulting model with the standard PINN, using the analytical inviscid solution as a reference. Although the artificial-viscosity model achieved a lower relative L^2 error, it produced a larger maximum pointwise error. This result prompted me to examine the shock location and transition width alongside aggregate error measures. The project strengthened my understanding that evaluating a numerical approximation requires considering local solution behavior and the physical assumptions introduced by the method, as well as its overall error.

Hybrid Finite Element–Neural Operator Solvers

During my research internship at Johns Hopkins University, I began developing a hybrid finite element–neural operator solver under the supervision of Professor Somdatta Goswami. The framework uses non-overlapping domain decomposition, with finite element and neural-operator subdomains exchanging information through a shared interface. I use GPU-accelerated JAX-FEM simulations to generate training data and train Transolver in JAX to learn the mapping from interface displacement fields to subdomain stress fields. The predicted stresses are used to compute interface reaction forces for the finite element solver, whose updated interface displacements become inputs to the next neural-operator evaluation. This coupling process helped me recognize that the neural operator's outputs influence the interface conditions it receives in subsequent iterations. Its approximation therefore needs to be assessed through its effect on interface compatibility, force balance, and the

convergence of the coupled system, beyond its performance on a fixed collection of samples. I also began to distinguish the speed of an individual neural-operator evaluation from the cost of obtaining a sufficiently accurate final solution. These considerations shaped my interest in how neural operators and numerical methods can complement one another, and how their integration should be evaluated as a complete computational method.

Because the interface displacements change during the coupling iterations, I began to consider whether the training data represented the conditions encountered during the actual solve. Using actual interface displacement data as a reference, I adjusted the generation of Gaussian random fields to produce interface displacement samples that better reflected the observed patterns. I then used JAX-FEM simulations to obtain the corresponding subdomain stress fields, forming a training dataset of paired interface displacement and stress fields. I also investigated convergence acceleration strategies and relaxation parameters, since the behavior of the coupled solver depends on how interface information is exchanged as well as on the neural operator itself. Through this work, I came to understand that training a neural operator requires anticipating how it will be used, while designing the solver requires understanding the conditions under which the operator can be relied upon. This perspective encouraged me to examine the training data, the coupling procedure, and the final solution together when evaluating the method.

Working with JAX-FEM and training Transolver in JAX has strengthened my interest in differentiable programming and simulation. I want to explore how neural operators and numerical methods can work together in computational workflows that connect physical modeling, simulation, and optimization. Within an end-to-end differentiable computational graph, material parameters, boundary conditions, and design variables can be connected to physical responses and objective functions, allowing sensitivities to be computed through automatic differentiation. This offers opportunities to use simulation for parameter identification, design optimization, and PDE control, where repeated evaluations and sensitivity calculations can be computationally demanding. I am interested in how the roles of neural operators and numerical solvers should be chosen for these tasks: which relationships are suitable to learn from data, which parts benefit from retaining an explicit numerical formulation, and how their combination affects efficiency and reliability. At the same time, a differentiable implementation does not by itself establish that its gradients faithfully represent the underlying physical model. I therefore want to understand how operator approximation, numerical discretization, and solver convergence affect the sensitivities used in optimization and control. More broadly, I hope to investigate how the requirements of a downstream task should guide the training, integration, and evaluation of neural operators, so that improvements in simulation translate into useful information for identifying parameters and making design or control decisions.

Collaboration and Open Research

Working with my teammates in the China International College Students' Innovation Competition helped me appreciate how discussion can improve our understanding of technical problems. We exchanged ideas about language-model fine-tuning, discussed unfamiliar tools, and worked through questions together, ultimately earning an award as a team. Explaining my reasoning and considering my teammates' perspectives helped me recognize gaps in my understanding and reconsider possible approaches. I also shared datasets and model checkpoints on Hugging Face and documented my research projects in public repositories. These experiences encouraged me to make my work accessible to others and to view communication as part of the research process. During doctoral study, I hope to contribute through thoughtful discussion, resource sharing, and collaboration, while continuing to learn from the experience and perspectives of others.

Future Work

Building on my research experiences, I hope to deepen my understanding of computational mechanics, neural operators, and differentiable programming and simulation, and explore how these areas can inform one another. I am interested in developing computational methods that support efficient physical simulation and its use in inverse problems, design optimization, and control. My current work provides a starting point for this exploration, while I remain open to methods and applications that may lead me to reconsider or extend my initial ideas. Through further study, experimentation, and collaboration, I hope to develop the judgment to identify meaningful research questions, choose appropriate approaches, and evaluate their contributions and limitations. My goal is to grow into an independent researcher who can take responsibility for this process and develop research directions grounded in both mathematical understanding and the needs of scientific and engineering problems.

In the longer term, I hope to pursue research and development in scientific computing, including opportunities in industrial research environments such as NVIDIA. I am drawn to work that connects advances in computational methods with the practical demands of scientific and engineering simulation. My experiences have made me interested in the path from an initial idea to a method that others can use: identifying the computational difficulty, developing an approach, examining its assumptions and limitations, and evaluating whether it offers a meaningful improvement within a larger workflow. I hope to take responsibility for this process as an independent researcher while working with colleagues whose expertise spans mathematics, mechanics, and computing. Doctoral study would give me the time and guidance to develop the theoretical depth and sustained research experience that this requires. Ultimately, I would like to contribute methods and tools that make demanding simulations more accessible and useful, while providing a clear account of their accuracy, computational cost, and range of applicability.